



In-depth astrosociological and physical analysis of the probability of the emergence of advanced civilizations and their interaction with humanity

This research report presents a comprehensive synthesis of cosmological, astrophysical, statistical and sociological data in order to map the probability of the emergence of the first intelligent civilizations in the universe and to assess the possibility of their physical or interdimensional presence in our planetary system. Given the extensive analysis of geospatial databases containing hundreds of thousands of valid records of unidentified anomalous phenomena (UAP) available in the web map https://svoboda-koduje.github.io/UAP_map/, it is necessary to subject this phenomenon to rigorous scientific examination. The visualization of this geospatial data, as represented by the iconography of map points with the symbol of a disk-shaped vessel, and their gradual depixelation and sharpening, metaphorically reflects the evolution of our understanding of the entire phenomenon. While in the past the data was blurry, ambiguous and often stigmatized, modern analytical tools, artificial intelligence and advanced sensors today allow us to see and categorize these objects with a high degree of physical and structural sharpness. This epistemological shift brings us to the point where anomalous objects are actually observed and measured in the context of operations in Earth orbit or during transmedia transfers in our atmosphere. The following analysis discusses fundamental questions about the origin of life, the limits of interstellar travel, and statistical patterns that may indicate systematic surveillance of our civilization.

Cosmological timeline and the formation of chemical prerequisites for life

Understanding the probability and time frame for the emergence of the first intelligent civilizations requires a detailed analysis of the cosmological timeline and the processes that led to the chemical enrichment of the universe. Based on current scientific consensus and measurements of cosmic microwave background radiation, the age of the universe is estimated to be approximately 13.8 billion years.¹ In the first fractions of a second after the Big Bang, the universe underwent a phase of cosmic inflation, during which it expanded exponentially at a rate exceeding the speed of light, leading to a homogenization of its structure.¹ This was followed by an era of primordial nucleosynthesis, which lasted from about the third to the twentieth minute of the universe's existence.² During this extremely hot and dense phase, only the lightest elements formed, with the matter of the universe consisting of approximately 75% hydrogen, 25% helium, and trace amounts of lithium and beryllium.²

For the next 380,000 years, the universe was an opaque plasma, with free electrons scattering photons. It was not until the temperature dropped to about 3,000 Kelvin that electrons could combine with atomic nuclei in a process called recombination, making the universe transparent and releasing the relic microwave noise we measure today.² This period was followed by the so-called Dark Ages (Ages), an epoch without any light sources that lasted several hundred million years.⁴ This period was completely devoid of heavier chemical elements (referred to as metals in astronomy), which completely ruled out the possibility of the formation of solid, rocky planets and the complex molecules necessary for biological life.

The first generation of stars and the role of nucleogenesis

The end of the Dark Ages occurred approximately 13.4 to 13.5 billion years ago (roughly 300 to 400 million years after the Big Bang), when the first stars, known as Population III, began to form from cold clouds of hydrogen and helium due to gravitational collapse.⁴ These stars were radically different from the stars we see today. They were theoretically extremely massive (hundreds of times the mass of the Sun), more luminous, and had very short lifetimes, measured only in units of millions of years.⁵ Because of their enormous mass, these stars underwent an accelerated cycle of nuclear fusion, during which they synthesized heavier elements such as carbon, oxygen, silicon, and iron in their cores. Their subsequent demise in the form of gigantic supernova explosions dispersed these newly formed heavy elements into the surrounding interstellar space.³ These first explosions forever chemically changed the universe and initiated the process of reionization.⁴

The material enriched by these first supernovae subsequently formed a second generation of stars (Population II), which are very low in metallicity but contain trace amounts of the elements necessary for the eventual formation of solids.⁸ Our own Sun, which formed about 4.6 billion years ago, or about 9.2 billion years after the Big Bang, is classified as a Population I star.¹⁰ This means that it represents at least the third (or later) generation of stars.⁸ The interstellar gas and dust from which our Solar System formed was recycled and enriched with material from many previous stellar cycles and supernova explosions over billions of years. This high level of metallicity (about 2% of heavy elements) was crucial for the formation of solid planetary cores, the formation of the Earth's crust, and the supply of biogenic elements (carbon, hydrogen, nitrogen, oxygen, phosphorus, sulfur - CHNOPS), which serve as the building blocks of biological life.¹⁰

Minimum volume of elements for the formation of planets and life

It did not take 9 billion years for the chemical elements to build up enough for terrestrial planets and life to form. Current astrophysical models and empirical data from exoplanet observations using satellites such as TESS and Kepler show that the process of solid planet formation is strongly dependent on the criticality of the metallicity in the protoplanetary disk.¹⁵ Research has set the threshold metallicity for the formation of Earth-sized and super-Earth planets at approximately one-tenth (0.1) to one-quarter (0.25) of the metallicity of our Sun.¹¹ Extremely metal-poor stars (with metallicities below -0.5 to -0.6 dex, or less than about 25% of the iron content of the Sun) have been shown to have a drastically reduced chance of forming massive rocky worlds, as they lack the necessary material for core accretion before the protoplanetary gas is dispersed by photoevaporation.¹⁶

However, the universe reached this critical threshold relatively quickly in selected regions. There is evidence that in some dense parts of galaxies, such as the globular cluster Messier 69, heavy element levels reached 22% of the solar level as early as 700 million years after the Big Bang.²⁰ Expert astrobiological studies extrapolate that the first rocky, potentially habitable planets could have formed in systems as old as 10 billion years, just 3 to 4 billion years after the Big Bang.²¹ This means that the universe had the necessary chemical and physical architecture for the emergence of biological life at least 5 to 6 billion

years before our planet even began to form.²² If we assume that the evolution of intelligence requires about the same time frame as on Earth (about 4 billion years), the first advanced technological civilizations could have existed in the universe when our solar system was just a cloud of cosmic dust. This time asymmetry opens up vast possibilities for the existence of super-civilizations with a technological lead of millions or billions of years.

The concept of the Galactic Habitable Zone and modern exoplanet mapping

The probability of the emergence and long-term maintenance of highly organized biological life is determined not only by the chemical composition of the parent star, but also by its precise location within the broader galactic morphology. Just as there is a circumstellar habitable zone around a particular star, there is also a concept in astrophysics called the Galactic Habitable Zone (Galactic Habitable Zone - GHZ).²³ This zone represents a ring-shaped region of the galaxy in which a sufficient degree of metallicity optimally intersects with an acceptable level of cosmic and radiation threat.²³

The density of stellar systems in an average spiral galaxy such as the Milky Way (which is estimated to be up to 13.6 billion years old) increases drastically towards the center.⁶ The galactic bulge and the immediate vicinity of the supermassive black hole Sagittarius A* are characterized by an extreme concentration of matter, leading to hyperactive star formation. This is closely related to the very high rate of massive star death in the form of supernovae.²⁴ Supernova explosions and gamma-ray bursts produce lethal doses of ionizing and X-ray radiation. If a potentially habitable planet were located too close to the center of the galaxy, it would undergo constant sterilization cycles that would repeatedly destroy any nascent ecosystems and strip away planetary atmospheres.¹³ Furthermore, the increased gravitational perturbation due to close passes of neighboring stars would make long-term stability of planetary orbits impossible, leading to extreme and deadly climate fluctuations.²⁴

At the opposite end of the galactic spectrum, at the outer edges of spiral arms and in the galactic halo, the situation is the opposite. Here, the density of stars is very low and the recycling of stellar material is not fast enough and frequent enough.²⁶ The result is a local metallicity that does not reach the necessary threshold values for the formation of massive terrestrial planets.²⁶ At the very edge of the galaxy, there is therefore not enough building material (silicon, iron, carbon) for the formation of solid worlds and the complex chemical compounds that condition the processes of life.

This implies that only the intermediate distance between the center and the edge of the galaxy is considered for smooth biological and civilizational evolution. In the case of the Milky Way, this Galactic Habitable Zone is defined as an annulus (ring) with an inner radius of about 7 kiloparsecs and an outer radius of about 9 to 10 kiloparsecs from the core.²³ Our Sun, located approximately 8 kiloparsecs (about 26,000 light-years) from the center on the inner edge of the Orion spiral arm, orbits in a nearly circular orbit that minimizes crossings of the dangerous spiral arms, thus providing our planet with the billions of years of environmental stability necessary for the evolution of complex organisms.¹³ Approximately 75% of the

stars in the same galactic habitable zone of the Milky Way are older than the Sun, which again strengthens the likelihood of the earlier emergence of alien intelligences.²⁷

Current status of exoplanet mapping (2025–2026)

Technological advances in astrophysics now allow us to empirically test these hypotheses. Starting with the discovery of the first exoplanet around a solar-type star (51 Pegasi b) 30 years ago in 1995, astronomy has witnessed a revolutionary development. As of 2025 and early 2026, NASA and ESA have catalogued more than 6,000 confirmed exoplanets through their observatories.¹¹ Many of these discoveries have been made using transit photometry (measuring the decrease in brightness of a star as a planet passes by) and radial velocity spectroscopy.

TESS (Transiting Exoplanet Survey Satellite), operating since 2018 and covering an area of sky 400 times larger than its predecessor Kepler, continues to systematically identify terrestrial planets, super-Earths, and gas giants across our galaxy.³⁰ TESS analyzes millions of stars and, combined with artificial intelligence algorithms such as the ExoMiner ++ model, verifies new planetary systems.³¹ In parallel, the European Space Agency (ESA) is using the CHEOPS satellite to provide highly accurate measurements of the radii of known exoplanets and to study their density and composition.³²

The most important breakthrough in current exoplanet mapping, however, is the move from mere detection to characterizing planetary atmospheres. The James Webb Space Telescope (JWST) is performing transmission spectroscopy, which identifies the presence of heavy elements, water vapor, and anomalous chemical imbalances (potential biosignatures) in the atmospheres of worlds tens of light years away.³³

This trend will culminate in the Habitable Project in the coming decades. Worlds Observatory (HWO), which is in active technological development in 2025–2026 as NASA's flagship mission for the 21st century. HWO is designed specifically to directly image Earth-sized exoplanets in the habitable zones of Sun-like stars and to routinely search for signs of a breathing biosphere and technosignatures.²⁹ All of this new data continues to confirm that Earth-sized planets are the statistical norm, not the anomaly, in the Galactic Habitable Zone.³⁷

Fermi Paradox, Drake Equation, and the Filtration of Intelligent Life

The fact that the universe has billions of stars in habitable zones, the necessary metallicity, and a time lead of billions of years leads to a fundamental scientific problem formulated by physicist Enrico Fermi in 1950. In a discussion with colleagues (Edward Teller, Emil Konopinski, Herbert York) at Los Alamos, he famously said: "So where is everyone?"³⁸ The Fermi paradox points to a stark contradiction between the mathematically extremely high probability of the existence of advanced extraterrestrial life and the apparent lack of indisputable physical, radio or engineering evidence of their activity (e.g. Dyson spheres, interstellar probes).³⁸

The fundamental analytical tool for grasping this paradox is the Drake equation. This equation was developed by astronomer Frank Drake and is a probabilistic model designed to estimate the number of active, communicating extraterrestrial civilizations (N) in the Milky Way.⁴⁰ The equation combines seven factors:

1. R^* : The average rate of formation of stars suitable for supporting life.
2. f_p : The fraction of these stars that have planetary systems.
3. n_e : The average number of planets in a system that have conditions suitable for life.
4. f_l : The proportion of planets on which life actually arises (abiogenesis).
5. f_i : The proportion of life-bearing planets on which intelligence evolves.
6. f_c : The proportion of intelligent civilizations that develop technology that releases detectable signals into space.
7. L : The average length of time (in years) that such civilizations have been transmitting signals or existing.

While the first three astronomical variables (R^* , f_p , n_e) are now determined with a relatively high degree of accuracy thanks to programs such as Kepler and TESS and their values are very optimistic, the biological and sociological factors (f_l , f_i , f_c , L) represent a huge unknown variable.³⁴ Depending on the settings of these unknowns, the result of the equation can take on values from a single civilization (us) to tens of millions of developed societies.³⁴ For a more sophisticated analysis of these probabilities, modern researchers use the tools of Bayesian statistics, which update the probabilities of hypotheses based on how long it took for life to emerge on Earth versus the emergence of intelligence, but taking into account the possible development of artificial intelligence, which can easily survive biological life and become the dominant form in the cosmos.³⁴

Grand Filter Hypothesis

The most convincing attempt to resolve the Fermi Paradox is the Great Filter hypothesis, first formulated by economist and visionary Robin Hanson in 1996.⁴² This theory assumes that there is one or more evolutionary barriers in the long chain of events that leads from the formation of the first molecules to an intergalactic expansive civilization capable of populating the galaxy (Kardashev scale types II and III). This step is either extremely statistically unlikely or fatally destructive.⁴²

The critical question remains whether the Great Filter is in our past or just ahead.⁴⁴ If the filter is in the past, it could have been the event of abiogenesis itself, the emergence of eukaryotic cells, the emergence of sexual reproduction, the development of the cerebral cortex, or the stabilization of climate through plate tectonics (catalyzed by the collision with the planet Theia).⁴³ If this hypothesis is correct, humanity is an incredible statistical anomaly and we are a rarity in the universe, if not entirely alone.

However, if the emergence of life and simple intelligence in the universe is relatively common (as suggested by the rapidity of the emergence of life on Earth after it cooled), then the much more chilling implication is that the Great Filter lies ahead. This would mean that the vast majority of technological civilizations are destroyed before they can expand to the surrounding stars. Fatal filters could include nuclear self-destruction (Fermi and Teller's original concern from the Manhattan Project), irreversible collapse of planetary climate, the threat of near-Earth objects (asteroids and comets), depletion of natural resources accompanied by massive demographic collapse (depopulation), or the creation of an uncontrollable artificial superintelligence (ASI) that eliminates its biological creators but has no need to colonize the entire universe in an expansive manner.⁴⁴

Zoo hypothesis and protection of local development

In addition to fatal filters, probabilistic models offer an explanation for the silence in the universe through the so-called Zoo hypothesis. This has been proposed to explain the absence of obvious technosignatures without necessarily assuming that intelligent extraterrestrial life does not exist or is bound to be destroyed.⁴⁷

The zoo hypothesis works on the assumption that extremely advanced entities (perhaps a pan-galactic federation or association of civilizations) have long known about the existence of humanity, but consciously and strictly avoid open interspecies contact.⁴⁶ Their motive is to allow us to develop naturally, socioculturally and technologically, without interference. Just as humans create nature reserves to protect endangered species without the disturbing influences of modern civilization, ETIs may keep our solar system in a kind of quarantine bubble, thereby simulating cosmic loneliness.⁴⁶ This hypothesis elegantly complements the empirical fact of occasional UAP incidents: highly advanced probes occasionally conduct data collection and monitoring, while rare and fleeting visual contacts occur, which do not have the mass and confirming character of an official landing.

Interstellar travel catalysts and advanced propulsion physics models

When considering whether extraterrestrial civilizations would ever invest enormous effort in visiting distant star systems, including Earth, we must factor into the research the astronomical fact that stars are finite.⁴⁸ No star has an inexhaustible supply of nuclear fuel. While smaller stars (red dwarfs) can live for trillions of years, medium-sized stars like our Sun, when they run out of hydrogen, expand into a red giant phase. This expansion will engulf the inner planets (Mercury, Venus, and most likely Earth) within a few billion years, effectively sterilizing any planetary system.⁴⁸ After the outer layers are shed, the star is left with a fading white dwarf.

Thus, if an advanced civilization has developed in a given solar system and has been able to overcome the Great Filters, it will eventually face a crushing physical reality: it must either accept the complete extinction of its own species along with its star, or expand into interstellar space.⁴⁹ This logically implies the evolutionary and existential necessity of learning to move between the stars. Survival of a species on

cosmic scales requires liberation from the shackles of a single biosphere and a single star. Any civilization old enough to face the decline of its homeland will inevitably become a civilization of stargazers.⁴⁹ Mastering propulsion capable of overcoming distances of light years becomes an imperative for the continuity of existence.

Warp drive physics (Alcubierre metric)

Standard Newtonian physics and Einstein's special theory of relativity set the absolute limit for the propagation of information and matter in the universe at the speed of light (c). For interstellar travel using conventional reactive propulsion (including fission or fusion rockets or solar sails), journeys would take tens to thousands of years, making them logistically extremely challenging for living biological beings.⁴⁹

However, advanced civilizations are likely to exploit deeper aspects of general relativity. In 1994, theoretical physicist Miguel Alcubierre proposed a mathematical model of the so-called Alcubierre drive . He found that general relativity does not prohibit the manipulation of spacetime itself. Instead of accelerating a ship through normal space, the Alcubierre drive would generate a force field that would contract the fabric of spacetime immediately in front of the ship while expanding it exponentially behind it.⁵⁰ The ship itself would remain within a safe local spacetime bubble (where space is flat) and travel at speeds exceeding the speed of light over cosmic distances without violating the rules of local relativity. A traditional problem with this model has been the requirement for vast amounts of so-called negative energy (exotic matter).⁵³

However, the latest scientific studies published in 2024 and 2025 in respected peer-reviewed journals (such as *Classical and Quantum Gravity*) have brought fundamental theoretical shifts. Researchers from organizations such as Applied Physics and Limitless Space Institute have shown that the physical model of warp drive can be restructured using positive energy, which obeys standard physics. By modifying the geometry of the bubble from a smooth ring to specific cylindrical segments, space manipulation can be achieved with much lower energy requirements, eliminating the need for fictitious negative energy.⁵⁵ Although current models of these “physical warp drives” (Physical Warp Drives) predict travel exclusively at subluminal speeds (below the speed of light), making science fiction concepts a real, if extremely difficult, engineering goal.⁵⁷ The ability of advanced civilizations, millions of years ahead in applied physics, to create these propulsion systems thus becomes a highly realistic scenario explaining the extreme flight characteristics of the phenomena observed in our atmosphere.

Multidimensional travel and the extraterrestrial hypothesis

In addition to linear travel in three spatial dimensions, modern cosmology and string theory (M-theory) introduce the possibility of additional, hidden dimensions. If the universe functions as a multidimensional multiverse or gateway, the UAP phenomenon may utilize the concept of interdimensional (multidimensional) travel.⁵⁹ This mechanism does not involve traveling through the interstellar void, but rather uses bridging from parallel spaces through points of contact with our physical reality, or through stabilized and traversable wormholes. wormholes).⁶⁰ This so-called Interdimensional Hypothesis (long advocated by French scientist Jacques Vallée) very elegantly

explains the highly strange properties of UAPs: their ability to materialize from nothing, instantly dematerialize, asymmetrically change shape, or pass through solid obstacles and absorb radar and light waves, which would appear like magic from our limited three-dimensional perspective.⁶¹

Another fascinating modification of space-time manipulation is the so-called Extraterrestrial model (Extraterrestrial Hypothesis). Biological anthropologist Dr. Michael P. Masters has formulated a radical idea based on evolutionary biology, hominid phylogeny, and theoretical physics: the entities operating within the UAP phenomenon are not beings from other stars, but represent our own, highly evolved human descendants from the distant future.⁶⁴ If humanity were to master the technology of creating closed time curves or retrocausal bubbles in the future, it could travel back in time. Masters argues that the anatomy of the most frequently described visitors ("Grays" – large brain, absence of hair, reduced jaws, distinct bipedal asymmetry) correlates precisely with long-term morphological trends in human evolution over the past millions of years.⁶⁴ Moreover, the travel of our descendants would be driven by an anthropological and rescue interest in exploring our own past (e.g., in times of global ecological upheavals).⁶⁴

Quantitative and morphological analysis of the UAP database: Identification of surveillance patterns

Under the strict assumption that advanced non-human intelligences (NHI) or other anomalous entities are visiting or monitoring our planet, geospatial analysis of historical reports provides invaluable hard data. The research and mapping report "Analysis and Prediction of the UAP Phenomenon", authored by Mgr. Jaroslav Svoboda, compiles a large, algorithmically merged database of **323,729 valid** UAP report records. This data comes from a synthesis of relevant global sources (including the American NUFORC, the French GEIPAN institute and the Czech Záře project) and maps incidents from the early 20th century to February 2026.⁶⁷ The database does not rely on anecdotal stories, but categorizes reports through an elaborate twelve-column architecture. It records the exact time, geodetic coordinates, shape (morphology), duration of the observation and applies natural language processing (NLP) to identify psychological interactions and described anomalous flight characteristics.⁶⁷

The data is divided into eleven specific "analytical missions," each tracking a specific observation profile. The following table shows the quantitative distribution of highly anomalous patterns within this vast dataset:

Analytical Category (Mission)	Total number of valid records	Year of peak activity	Dominant morphological shape
Triangles	20,940	2012	Triangle
Silent flight	12,444	2012	Triangle
Extreme speed	11,881	2012	Light
Tic-Tac / Cigar / Cylinder	11,881	2012	Cigar / Top Hat
Water worlds (proximity to water)	9,055	2012	Light
Aerial observations	8,476	2013	Disk
Military / Radar detection	3,955	1952	Disk
Nuclear footprint	1,501	1967	Disk
Mass observations	953	1966	Disk
Psychological impact	477	1977	Disk
Trusted witness	422	1952	Light

Morphological evolution and the phenomenon of confirmation bias

One of the most fundamental findings of the research is the revelation of a diametrical difference between the sightings in the historical era (1950s to 1970s) and in the modern era. The historical period was absolutely dominantly associated with the shape of the “Disc”.⁶⁷ However, the analysis revealed that the “flying saucer” narrative was largely contaminated by a structural journalistic error. In 1947, during a formative sighting, pilot Kenneth Arnold described the unknown craft more as crescents or bat-wing-shaped objects, but to illustrate their *flight dynamics*, he said that they moved as if you were “throwing a saucer across the water”. The press coined the acronym “flying saucer” saucer”. This media construct subsequently caused enormous conformational distortion and social contagion in data for decades to come, as people unknowingly projected this shape onto any obscure point in the sky.⁶⁷

The true nature of the objects would emerge in the data in later decades. During the 1980s and 1990s, reports of massive, silent triangles and chevrons, often the size of several football fields, dominated the reports. Although this wave coincided with the testing and deployment of top-secret stealth bombers (such as the F-117 Nighthawk and B-2 Spirit), analysis of the hard data shows that the behavior of the reported UAPs defied any human thermodynamics. These objects were capable of hovering absolutely still for hours over a single location without any aerodynamic engine noise, and then applying what is known as an “unearthly instantaneous acceleration” into the stratosphere.⁶⁷ These 12,444 reports of silent flight and extreme maneuverability cannot be attributed to any existing conventional aircraft using lift.

In the modern dataset, culminating around 2012–2013, the category of cylindrical or oval bodies without wings and control surfaces, made famous by the media after the USS Nimitz aircraft carrier fighter incident (2004) as the so-called Tic-Tac, dominates. The database contains a staggering **11,881** valid records of this shape between 1901 and 2026. A deep analysis of this data reveals subcategorizations including 7,724 cases with a cigar morphology (Cigar), 4,071 cylinders (Cylinder), and 22 capsules.⁶⁷ The key piece of evidence that rules out skeptical arguments about mere light hallucinations is the symmetry of the time of observation: of these nearly 12,000 cases, a full 5,859 were observed at night and 5,260 were recorded and described in full daylight, indicating the existence of solid, physical objects.⁶⁷

Surveillance patterns and statistical probability of visits

The long-term pattern of extraction from the database reveals an evident focus of the UAP phenomenon on infrastructure associated with the military and technological dominance of humanity. A clear surveillance operation can be identified. The “Nuclear Footprint” analysis points to 1,501 precisely geolocated incidents that took place in the immediate vicinity of nuclear power plants and military forces with intercontinental missiles (with a historical peak in 1967).⁶⁷ This is interspersed with 3,955 cases recorded by military units and radars (Military/Radar), the peak of which falls on the early phase of the Cold War and the famous summer of 1952.⁶⁷

During the so-called Washington Flap in July 1952, unidentified objects (bright orange spheres exhibiting absolute flight independence from inertia) violated the restricted airspace over the Capitol and the White

House. The analysis notes that this was an extremely validated case, in which the objects not only changed speeds beyond the capabilities of the Air Force at the time, but were tracked by up to three independent radar systems (National Airport and Andrews Air Force Base) and simultaneously confirmed by experienced visual observers from airline and military interceptor pilots (F-94).⁶⁷ Cross-validation of the radars excludes atmospheric inversions. Other historical files in the database clearly document the monitoring of strategic locations, such as the pursuit of military targets by fighter flights in 1949 and 1960, the disruption of the electric field over nuclear weapons, and interactions with electrical power grids on the ground, suggesting an effort to test the limits of human understanding of electromagnetism.⁶⁷

When calculating the probability and extent of the phenomenon, we have to work with filtered hard data. The sum of the highly relevant and difficult to question security categories of the database, i.e. 3,955 records (Military/Radar) and 1,501 (Nuclear Track), gives approximately 5,456 cases of deep surveillance . If we add to these 8,476 highly credible cases observed by experienced pilots in both commercial and military aviation, we get a sample of almost 14,000 high-profile unexplained events.⁶⁷ Out of a global database of 323,729 reports, these completely credible, technologically anomalous and radar-supported cases represent approximately **4.3%** .⁶⁷ However, if we include all cases from the “high strangeness” section (High Strangeness), which includes records objectively describing parameters beyond the physics of conventional aviation – such as 11,881 reports of speeds exceeding the tremors of hypersonics , 12,444 observations of completely silent platforms without lifting surfaces, and 9,055 cases of transmedia objects capable of moving from the atmosphere deep into the world’s oceans (“Water Worlds”) – we get a proportion of cases that clearly indicate the operations of technologically superior extraterrestrial or interdimensional infrastructure, reaching approximately **10% to 15%** of the entire analyzed sample of reports.⁶⁷ This fraction of the data represents a fundamental phenomenon that cannot be explained by errors or secret government projects.

The visually interactive system itself, for which new programming rules are being created within the framework of Turf.js and Postgres with geolocation (PostGIS), seeks to utilize these structures. Spatio-temporal clustering extraction (spatiotemporal clustering) and correlation graphs will allow the construction of probabilistic neural networks and the deployment of machine learning algorithms that should proactively predict the occurrence and tracking of other UAPs.⁶⁷ The database and interactive map thus serve not only as a chronicle of phenomena, but as an exact predictive model of the behavior patterns of alien intelligence.

Institutional and academic legitimization of anomalous phenomena research

Although for decades UAP research faced a strong scientific and governmental taboo ,⁶⁸ the last years of the 21st century have seen a massive epistemological shift. The topic is no longer seen as a marginal pseudoscientific fascination, but has come under the wing of serious academic and governmental research bodies.⁶⁷ The emergence of government offices such as the US AARO (All- domain Anomalies Res-

olution Office), and the involvement of organizations led by NASA demonstrate efforts to apply exact science and avoid security risks over protected facilities.⁷⁰ Three key entities can be highlighted from the university and technological environment:

1. The Galileo Project (Harvard University): A project led by astrophysics professor Avi Loeb at Harvard represents the absolute pinnacle of efforts for transparent and validated systematic research into physical extraterrestrials . technosignatures near Earth. Unlike classical SETI , which searches for radio signals deep in space, Galileo is tasked with detecting physical objects and remnants of extraterrestrial technology.⁷² The project has built independent observatories across the US (including Massachusetts and Pennsylvania) with advanced multispectral sensors (infrared, radio, sound and optical surveillance). The huge data series captured is continuously analyzed by AI algorithms to distinguish elements of human activity from objects originating from outside the Earth.⁷² The project is also preparing proposals for space missions (rendezvous) to study interstellar bodies with anomalous properties, similar to the famous object 'Oumuamua.⁷³

2. SETI Institute & Berkeley SOWING Research Center: A historic and respected institution that continues to innovate. Previously dependent primarily on the analysis of standard narrowband radio signals (using infrastructure such as the Allen Telescope Array and engagement of SETI @home software for citizen scientists⁷⁶), applies Murchison networks in 2025 Widefield Array for detecting low-frequency technosignatures across thousands of other galaxies. In addition, SETI experts are analyzing the extent to which alien civilizations would recognize us due to our atmospheric pollution, and are looking for signs of advanced engineered macrostructures, such as Dyson spheres.⁷⁷

3. University of Würzburg (IFEX): Interdisciplinary Center for Extraterrestrial Studies (IFEX) in Germany, led by Prof. Hakan Kayale , has developed a revolutionary stance in the field of European UAP research.⁷⁹ Kayal's team is building intelligent sky camera and sensor stations using artificial intelligence and machine learning to automatically register objectively unexplained anomalies.⁸⁰ In addition, IFEX brings together experts by organizing regular fall scientific conferences on the integration of UAP science research into mainstream physics, and calls for strong linkages between government flight procedures and public reporting of eyewitness sightings by airline pilots.⁷⁹

These systems are slowly and surely breaking down age-old taboos and applying exact instrumental analyses to clarify the phenomenon and obtain valid correlations.

Astrosociological models of contact and psychological assimilation of the phenomenon

Let us now try to apply the predictive model to what our hypothetical future or even ongoing contact with a potential advanced civilization might look like. The idea of an extraterrestrial encounter is not just a matter of physics and engineering, it is primarily a fundamental question of existence, psychology and theology.⁸¹

The unreality of political and governmental "Landing on the lawn"

The expected scenario from the movies, where a spaceship descends over the United Nations building and representatives of an advanced galactic civilization meet with representatives of the governments of the United States or other great powers, is deeply illogical under analytical scrutiny. Our humanity is fragmented; there is no unified planetary structure capable of speaking for our entire species. Human nature is largely built on tribalism, territorial wars, and strong national prejudices.⁶⁹

As defined in an extensive Brookings analysis Institution, called the Brookings Report in 1960, commissioned by NASA, there are immense risks arising from the discovery of alien intelligence. The report concluded that any uncontrolled disclosure of confirmed alien beings or their technology could lead to the absolute collapse of society. This phenomenon, called "ontological shock", could disrupt traditional institutions of power. Religious models, especially the Abrahamic faith, which bases human existence on the absolute election of God (although even theologians today find room for integrating Christianity and the discovery of other forms of life), would have to undergo a stormy reevaluation, and radical groups could interpret the phenomenon as apocalyptic or as a manipulation by demonic forces.⁶⁹

Moreover, the official government apparatus views everything through the lens of the military "threat paradigm." An entity that has ships of unknown physical properties breaching the world's airspace without the possibility of any human defense represents a theoretical threat to the security of the state.⁶⁹ For a highly advanced interstellar or interdimensional entity, official contact would be counterproductive, as it would automatically provoke a panicked defensive and probably even aggressive militaristic reaction out of fear and loss of the power ego.

Control System Theory and the Soft Disclosure Phenomenon

A more realistic and likely model of ongoing contact corresponds to the so-called Control System Hypothesis (Control System Hypothesis). System Hypothesis, developed by Jacques Vallée.⁸⁴ Vallée believes that the UAP phenomenon is not a program of classical interplanetary diplomatic rapprochement. Rather, it acts as a mechanism of cybernetic control—a delicate planetary thermostat of human consciousness.⁸⁴ Whenever human belief systems and dogmas (whether scientific or religious) become rigidly limited, the phenomenon manifests itself with a high degree of absurdity, pushing the boundaries of our understanding, thereby ensuring our cognitive and cultural flexibility in the long term.⁸⁴ These operations do not

respect state or defense protocols, but often bypass elites and speak to individual witnesses in the context of the symbolism of the current era.⁸⁶

For this reason, another scenario of continuous contacts is being fulfilled: the so-called **Soft Disclosure**, or smooth, non-violent and gradual disclosure through disruption and change of the entire cultural field.

⁸⁷ Sociologists have long studied the viral absorption of the UFO phenomenon into global pop culture. For decades, society has been subconsciously and systematically educated and dumbed down through science fiction series (*The X-Files* , *Star Trek*), billion-dollar Hollywood spectacles (*Close Encounters of the Third Kind* , *The Incoming*) and literature.⁸⁸

Pop culture serves here not only as entertainment, but as a gigantic tool of mass psychological conditioning (cultural conditioning).⁹⁰ It shapes the narrative of “Gods from the Stars,” but with the secular and technological rhetoric of our present. This conditioning has an extremely effective dampening effect: when in 2025 governments finally declassify information through the testimony of Air Force whistleblowers that we are dealing with asymmetric drones of non-human origin operating in the Earth’s atmosphere, it will not provoke the planet-wide anarchy and destruction predicted by the Brookings report in the 1960s.⁷⁰ The majority of the population approached the phenomenon with stoic calm and a simple shrug of the shoulders, since the cognitive integration (assimilation) of unknown entities into human reality had already been mediated in the form of simulation through pop culture myths.⁹⁰

Moreover, the personal level of contact moves from passive fear-sharing (analyzing abductees into alien ships) to citizen initiative. There are communities around the world conducting meditation sessions known as CE-5 (Close Encounters of the Fifth Kind) protocols that integrate the science of human consciousness, quantum connectivity, and technology in a unique effort to influence the arrival of UAPs.⁹² This two-way fluid intersection through culture, spirituality, pop culture, and the gradual acceptance of science thus demonstrates an organic assimilation of the phenomenon that an aggressive formal political act could never achieve without collateral damage to the civilizational format.

Final synthesis

The emergence and detection of advanced civilizations represent a complex interweaving of disciplines at the interface of astrophysics, geography, history, technology and philosophy. Our knowledge of nucleogenesis and the age of the universe confirms that the galactic system reached chemical complexity (high metallicity) sufficient to create terrestrial ecosystems billions of years before the formation of the Earth. The location of our blue planet in the central Galactic habitable zone and protected developmental cushion indicates that any civilization that has overcome the feared Great Filters examined in the Fermi Paradox will be asymmetrically advanced towards humanity, using technological manipulation of gravity, positive energy within the warp field or mastering the physics of parallel dimensions.

Geospatial analysis of historical UAP databases fully confirms traces of operational surveillance with unquestionable peaks above the nuclear triad infrastructure and radar points of military powers. At least a tenth of the recorded reports out of a total of 323,729 cases are characterized by highly strange flight

characteristics that destroy the current concept of aerodynamics and do not exclude constant observation or surveillance from the position of superior systems (including the Zoo hypothesis). The absence of full-fledged diplomatic contact does not stem from its non-existence, but comes from the necessity to avoid the ontological and existential shock of our power irrelevance. While scientific discourse through subjects at Harvard, in SETI and IFEX turns age-old social taboos into hard empirical detection, soft disclosure realized through the assimilation of space beings in pop culture fulfills the purpose of a training simulator, protecting the psyche and social stability of humanity during its gradual integration into a much wider cosmic domain.

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